Experiment: Logical, Bit-Manipulation and Shift Operations Using 8-bit and 16-bit Microprocessors

1. Aim

To write and execute assembly language programs to perform logical, bit-manipulation, and shift operations using 8085 (8-bit) and 8086 (16-bit) microprocessors.

2. Objectives

1. To perform logical operations such as AND, OR, XOR, and NOT.
2. To understand bit manipulation using rotate and complement instructions.
3. To perform shift operations and verify the results.

3. 8085 Assembly Language Programs (8-bit)

A. AND Operation

```
LDA 2050H
MOV B,A
LDA 2051H
ANA B
STA 2052H
HLT
```

B. OR Operation

```
LDA 2050H
MOV B,A
LDA 2051H
ORA B
STA 2052H
HLT
```

C. XOR Operation

```
LDA 2050H
MOV B,A
LDA 2051H
XRA B
STA 2052H
HLT
```

D. NOT (Complement) Operation

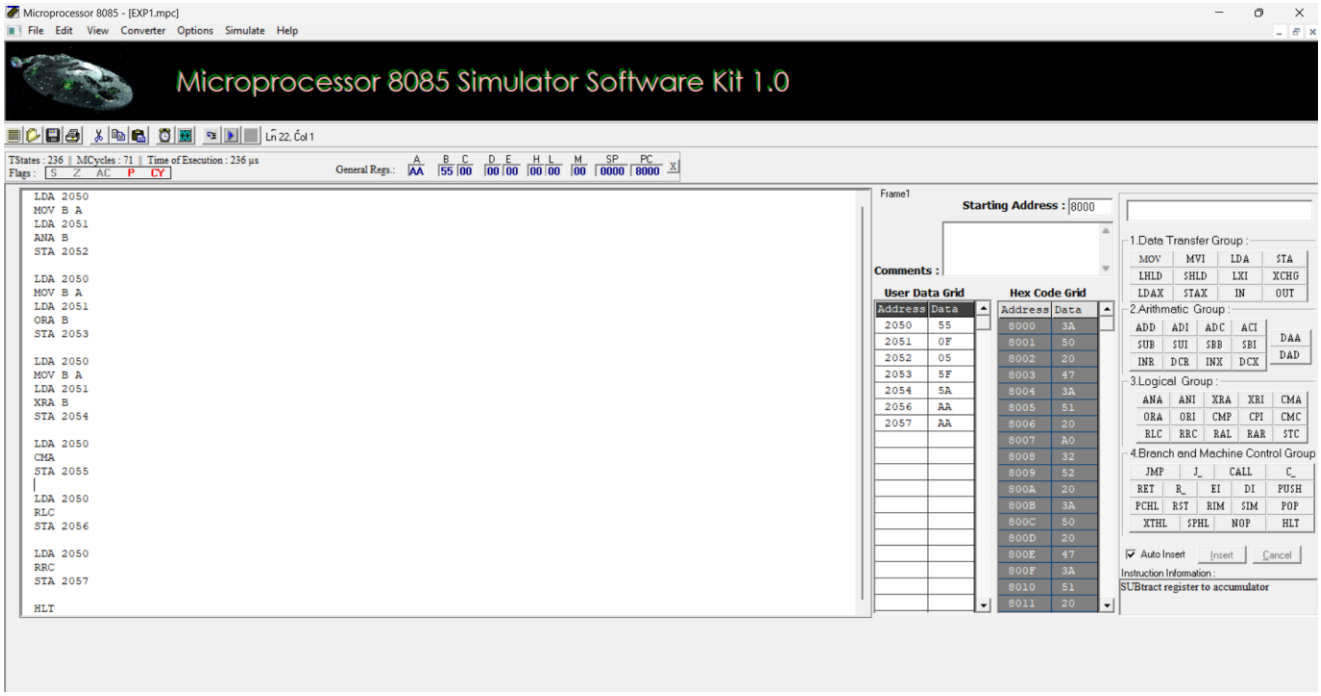
```
LDA 2050H
CMA
STA 2052H
HLT
```

E. Rotate Left

```
LDA 2050H
RLC
STA 2052H
HLT
```

F. Rotate Right

```
LDA 2050H
RRC
STA 2052H
HLT
```



6. Result

Logical, bit-manipulation, and shift operations were performed using 8085 and 8086 assembly language instructions, and the results were verified using the respective microprocessor simulators.